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STANDARD FOR T320HVN TFT-LCD SKD (A Grade)

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AU OPTRONICS CORPORATION

Specification for Approval

INCOMING INSPECTION STANDARD FOR T320HVN A Grade TFT-LCD SKD (open cell)

The content of this technical information
is subject to change without notice.
Please contact AUO or its agent for
further information.

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Customer	Checked & Approved by
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1 Scope:

- 1.1 The incoming inspection standards shall be applied to TFT-LCD SKD (hereinafter called "SKD") that supplied by AU Optronics Corporation (hereinafter called "seller").
- 1.2 Specifications contains
 - Electrical inspection specification
 - Appearance specification
 - Outside dimension specification

2 Incoming inspection:

The buyer (customer) shall inspect the SKDs within twenty calendar days of the delivery date (the "inspection period") at its own cost. The results of the inspection (acceptance or rejection) shall be recorded in writing, and a copy of this writing will be promptly sent to the seller.

Should the buyer fail to notify the seller within the inspection period, the buyer's right to reject the SKDs shall be lapsed and the SKDs shall be deemed to have been accepted by the buyer.

3 Inspection instruments:

- 3.1 Pattern generator: PCB Tester or equivalent model.
- 3.2 Video board: AUO video board or equivalent. The output of the signal should comply with the specification provided by AUO.
- 3.3 Luminance colorimeter: Topcon BM-7 or equivalent model

4 Inspection environment conditions:

- 4.1 Room temperature: 20 ~ 25°C.
- 4.2 Humidity: 65±5% RH.
- 4.3 Illumination: Fluorescent light (Day-Light Type) display surface illumination to be 300 ~ 700 Lux. (standard 500Lux.)
- 4.4 To be a distance about 50±5 cm in front of LCD unit, viewing line should be perpendicular to the surface of the SKD judge the visual appearance with human's eyes
- 4.5 Take off the protector of polarizer while judging the display area.
- 4.6 If there is any question while judging, check the panel again while operating.

5 Inspection Criteria

- Electrical inspection specification

5.1 Dot Defect

- A. Every dot herein means a Sub-Pixel (each Red, Green or Blue color).
- B. Dot defect is defined as that the defective area of the dot is larger than 50% of the dot area

5.1-1 Bright Dot

Bright Dot is defined as Dot (sub-pixel), which appears bright on the screen when the LCD module displayed at black pattern.

Inspection Item	Criteria
R.G or B 1 dot	Max 0 allowed

A partial bright dot damaged less than and equal half size of sub-pixel is not counted as a bright dot defect and should be specified below.

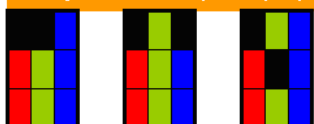
Inspection Item	Criteria
5% ND-Filter Not-visible	Ignored
5% ND-Filter Visible 2% ND-Filter Not-visible	Max. 6 allowed
2% ND-Filter Visible	Max. 5 allowed

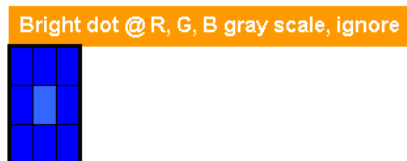
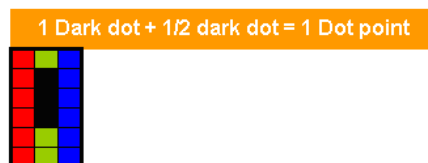
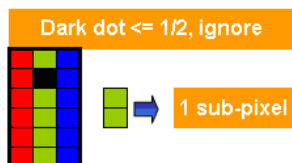
5.1-2 Dark Dot

Dark Dot is defined as Dot (sub-pixel), which appears dark on the screen when the LCD Module displays at bright pattern.

Inspection Item	Criteria
R.G or B 1 dot	Max. 8 allowed
Adjacent 2 dots	1 pairs
Adjacent 3 dots	0 pairs

Adjacent 2 dots (example)





5.1-3

Total amount of Full Dot Defects	Max. 8 allowed
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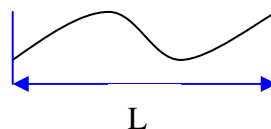
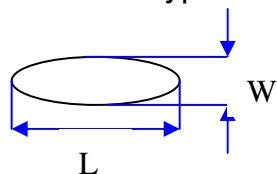
5.2 Appearance inspection specification

Judge area	Judge item		Inspection specification		
	Polarizer Defect (can't remove)	Round	Average diameter :(mm)		Numbers
			D<0.5		Disregarded
			0.5≤D≤1.3		N≤5
		1.3<D		N=0	
		Liner	Width: W (mm)	Length: L (mm)	Numbers
	W<0.15		L<10	Disregarded	
	0.15≤W≤0.3		10≤L≤30	N≤5	
	Dirt (can't remove)		As above		
	Grains				
Glass broken					
Bubble in cell (Active area)		D≤0.5 mm , N≤4			
Silicone	Silicone spread		The height can't over polarizer (Fig.1)		
	Silicone residue		Residue on FPC or PCB ≤2mm (Fig.1)		
Glass	Wire (on array)		No damaged≤1/3 lead length of efficient circuit		
	Edge	No extended crack,			
		Thickness of broken area: T≤0.5mm (Fig.2)			
		Width of broken area: don't care (A of Fig.3)			
		Length of broken area: ≤4.0mm (D of Fig.3)			
		Width from BM to broken edge: ≥2mm (B of Fig.3)			
		Protrusion (Burr) of glass: ≤0.3mm (C, E parts of Fig.3)			
The edge of active area	The end of polarizer		The width from the edge of polarizer to the edge of inner BM can't be less than 1.6mm. (Fig.4)		
PCBA	Appearance		Can't see the abnormal color and expose the cupper of PCB		
	Component		No damage		
Connector	Connection status		Need correct connection.		
FPC/FFC	Broken		No		

Note 1: It can't count the appearance defect if it can be clean, such as extraneous substances which can be wiped out, like Finger Print, Particles are not considered as a defect.

When $L \geq 2W$, defect count as liner type.

Note 2: $D = 1/2(W + L)$



Note 3: Defects which is on the Black Matrix (outside of Active Area) are not considered as a defect.

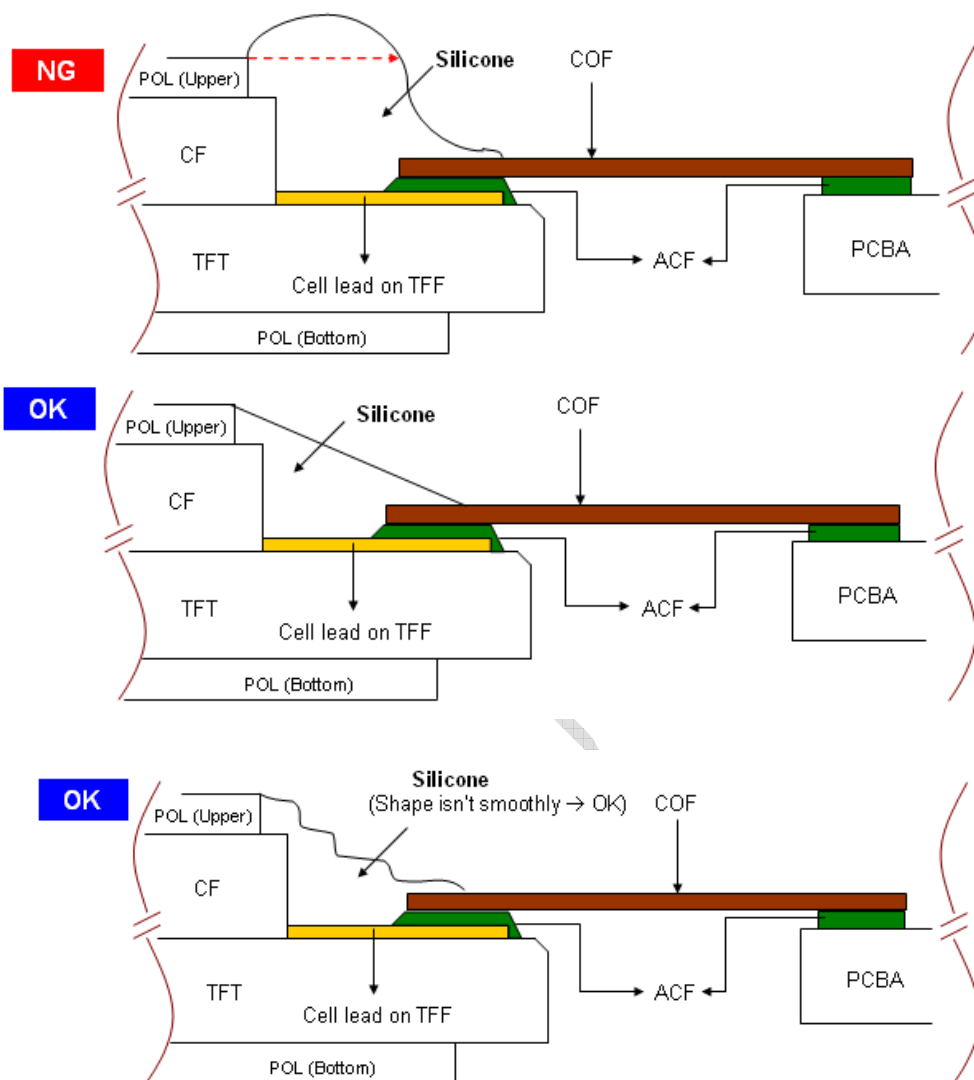


Fig.1

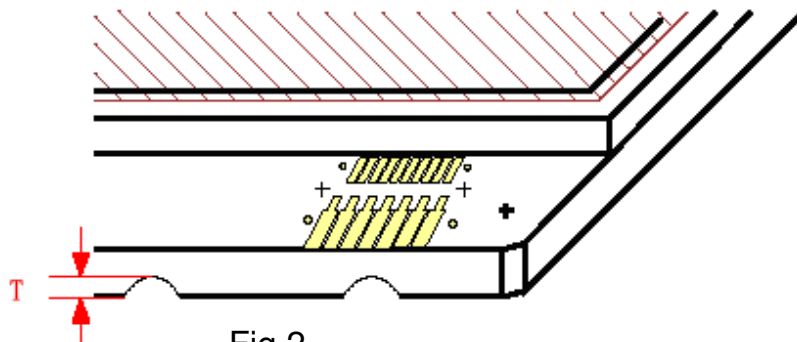


Fig.2

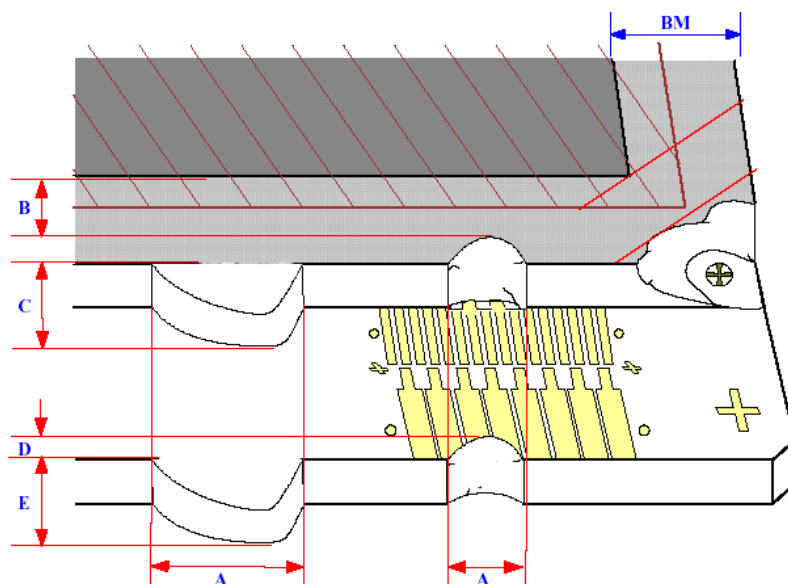


Fig.3

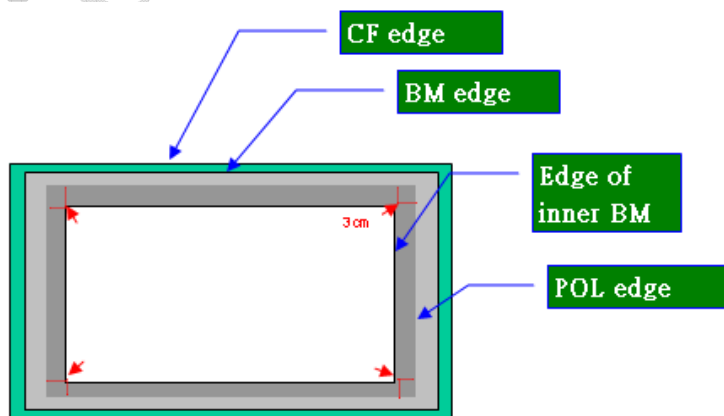


Fig.4

5.3 Mura

All kinds of Mura should be invisible at L128 pattern while 2% ND filter applied.

Movable Mura should be ignored if the Mura disappears after lightly patting on the Mura area.

And the line defect can be counted as Mura which can be invisible while ND2% applied.

6 Inspection judgement:

If the protection film of polarizer in TFT side is already removed (not remove area is less than palm area), and there are some abnormality on product appearance (for example: polarizer scratch...), buyer should take responsible for this and can't do RMA return.

7 Precaution:

Please pay attention to the following items when you use the LCD without back-light unit.

- 7.1 Do not twist or bend the SKD and prevent the unsuitable external force for display SKD during assembly.
- 7.2 SKDs taken out from the box should be friendly to avoid damaging COF lead .
- 7.3 Operator is forbidden to use hands to handle SKDs. Vacuum pad & appropriate vacuum pressure should be applied for the handling process. Customer need to take responsibility if PCB or COF suffer damage by customer induce
- 7.4 Make sure of operator wearing Wrist Strap during any SKD handling process for ESD protection.
- 7.5 Static electricity will damage the SKDs; please do not touch the SKD without any grounded device and always keep static electricity under 1KV when handing or assemblage.
- 7.6 Management Method of Ionizer: Decay time : < 1KV within 5 sec & balance : +/-100V or less.
- 7.7 Adopt measures for good heat radiation. Be sure to use the SKD with in the specified temperature.
- 7.8 Avoid dust or oil mist during assembly.
- 7.9 Follow the correct power sequence while operating. Do not apply the invalid signal, otherwise, it will cause improper shut down and damage the SKD.
- 7.10 Less EMI: it will be more safety and less noise.
- 7.11 Please operate SKD in suitable temperature. The response time & brightness will drift by different temperature.
- 7.12 Avoid displaying a fixed pattern (exclude the white pattern) for a long period, which may lead to image sticking.
- 7.13 Be sure to turn off the power when connecting or disconnecting the circuit.
- 7.14 Polarizer scratches easily, please handle it carefully.
- 7.15 Display surface never likes dirt or stains.
- 7.16 A dewdrop may lead to destruction. Please wipe off any moisture before using SKD.
- 7.17 Sudden temperature changes cause condensation, and it will cause polarizer damaged.
- 7.18 High temperature and humidity may degrade performance. Please do not expose the SKD to the direct sunlight and so on.
- 7.19 Acetic acid or chlorine compounds are not friends with TFT display SKD.
- 7.20 No parts of the panel should be dismantled without the presence of AU engineer or the

product will not be warranted.

- 7.21 No strong vibration or shock. It will cause SKD broken. And the glass broken will count to customer's responsibility passed through customer the first working station.
- 7.22 Be careful of injury from a broken display SKD. Please avoid the pressure adding to the surface (front or rear side) of SKDs, because it will cause the display non-uniformity or other cosmetic and functional defect.
- 7.23 All defects are caused by customer induce , customer should take the responsibility
- 7.24 LCDs contain a small amount of Liquid Crystal. Please follow local ordinances or regulations for disposal.